Episcaf v3: designing from an energetic definition of the epitope

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Living record. Read top to bottom to get up to speed. Every methodological choice is a dated entry in `docs/DECISIONS.md`.

This is the running record of a new design avenue for episcaf, kept as its own subproject (`episcaf_v3/`) inside the episcaf repository. The shipped method at the repository root ("v2") scaffolds antibody epitopes defined by 3D *geometry*; this log asks whether defining the epitope by *energy* — which residues actually carry the binding — and letting the design constraints follow the energetics produces better scaffolds. It is built one recorded decision at a time (`docs/DECISIONS.md`) and is meant to be read in order: each section is one iteration of what we tried, what we found, and what it changed for the next step. It is not a formal manuscript, and the method here is early — most of it is still to build.

1 From geometric to energetic epitopes

1.1 What the previous method did

The shipped method ("v2"; it lives at the root of this repository, with `episcaf_v3/` as one subdirectory beside it) scaffolds *B-cell epitopes* — the regions of an antigen an antibody recognizes — so that a high-throughput serology assay (PepSeq) can test conformational, discontinuous epitopes that linear peptide tiling cannot represent. The epitope is taken from an antibody:antigen crystal structure as the set of antigen residues the antibody contacts, defined by a heavy-atom distance cutoff. Those residues are embedded into a *de novo* protein with RFdiffusion, given a sequence by ProteinMPNN, and folded by AlphaFold3 to check whether the design actually holds the epitope in its native geometry. Designs are kept or rejected on four filters: the epitope and overall backbone RMSD to the intended geometry, the predicted alignment error, and whether scaffold mass clashes into the antibody's binding path. Where an antigen has no solved antibody, a native-aware cylinder surrogate stands in for that clash filter. A composite scorer then ranks the survivors and picks the best designs per epitope. This pipeline produced DP4, a 36 000-member PepSeq library now in synthesis.

1.2 Where the geometric definition runs out

The method works, but two problems trace to the same root: it treats the epitope as a flat set of geometric contacts, all equally important.

The first is that binding energy is not uniform across an interface. A handful of "hot-spot" residues typically carry most of the binding free energy, surrounded by a ring of residues that contribute little directly but shape the pocket and exclude solvent. A geometric cutoff cannot see this: it labels every contacted residue an epitope residue, and the contig we hand RFdiffusion then fixes the identity and backbone of all of them equally. We are constraining residues we have no evidence contribute to binding, which costs the generator design freedom it may need to fold a good scaffold — and we do not know which contacts are carrying the binding.

The second is visible in the hard cases. A contig enforces epitope residue identity and ordering, but it does not strongly constrain the distances and orientations *between* spatially separated epitope islands. So the model can satisfy the contig and still lose the geometry an antibody needs. The clearest symptom is PDB `2XQB` patch `OP`, a 56-residue epitope split across two islands: across 2624 RFdiffusion trials, zero designs passed the filters. A 0% rate over thousands of attempts is a real failure of what the contig constrains, not undersampling.

Underneath both is a definitional gap. "Epitope" is loosely and inconsistently defined in the literature — usually by a geometric contact convention chosen for convenience — and that ambiguity propagates straight into the design constraints. If we cannot say which residues matter and why, we cannot say which ones a design is actually required to reproduce.

1.3 The new avenue: define the epitope by energy

v3 replaces the geometric definition with an energetic one. Instead of asking which residues *touch* the antibody, we ask which residues *carry the binding*, and we let the design constraints follow the energetics. Concretely, we sort the interface into three categories and constrain each according to what its job is:

- **Category 1 — energetic (hot-spot) residues.** The residues that dominate the binding free energy. These retain both their *exact identity and their shape*: the contig fixes their backbone and their sequence is locked to native. This is what we currently do to the whole geometric epitope; here we do it only where a residue carries the binding.
- **Category 2 — structural (occluding) residues.** The rest of the contact shell. These matter for how the site fits together — they hold the shape and exclude solvent around the hot spots — but their identity is not the point. So we fix their *shape* (the contig holds the backbone) but let ProteinMPNN choose the sequence. Their correct orientation can be enforced by superimposing them on the native antigen and filtering on a small RMSD.
- **Category 3 — the rest of the scaffold.** These residues exist to enforce the geometry of Categories 1 and 2. They can be any identity and adopt any structure, subject to two constraints: they must not introduce new steric clashes with the antibody, and they must not compete with the binding site. Everything else about them is free — scaffold length and the position of the epitope within the construct are randomized.

By fixing identity only where it carries the binding, and shape only where a residue's job is structural, we give the generator more room to build a foldable scaffold while still holding the residues that bind. The same energetic reading also sharpens the hard-case problem: it tells us which inter-residue geometry is worth enforcing, rather than enforcing all of it blindly.

A second change makes the pipeline cheaper and is, in hindsight, overdue: we can filter RFdiffusion backbones *before* ProteinMPNN and AlphaFold3. A backbone that already places the epitope in the wrong orientation, or drapes scaffold mass through the antibody's path, was never going to produce a passing design; superimposing the fixed epitope on the native antigen and checking a small RMSD plus an antibody-clash test at the backbone stage discards those doomed designs before we spend the expensive sequence-design and folding steps on them. This is the same accessibility-and-fidelity check the shipped scorer applies at the end, moved to the front.

1.4 What this rests on

The approach depends on getting the energetics right, and the hard part is solvent. A naive pairwise antigen–antibody interaction energy will mislead: it ignores the desolvation a residue pays to make a contact, and it misses contributions mediated by ordered water — a bridging hydrogen bond can be load-bearing yet invisible to a direct pairwise sum. Deciding how to compute a defensible per-residue energy, including solvent, is the first real decision of v3, and it is where the record begins.

We do not take the ranking on trust. The first step is to reproduce known numbers: take a complex with experimentally measured binding energetics and check that our per-residue energies recover its known hot spots before we let the definition drive any design. The DP4 binding data, when it returns, gives a second, independent test — whether designs that better preserve the energetic residues bind better.

We are not starting the energy machinery from scratch. The sibling `bcell_epitope` project already computes a per-residue interface interaction-energy decomposition — including a water-mediated bridge channel — on a documented GROMACS stack, and it is the template we build the energy engine from. We treat it as a reference, not a protocol to inherit: its "epitopeness" is an interaction energy, deliberately not a binding free energy (it omits desolvation and entropy), so whether v3 adopts that quantity or a fuller one is itself a decision to make and record, not to assume.

1.5 Status

The method above is a plan, not a result: nothing here has been built or tested yet. What follows in this log will be the decisions, one at a time — the energy method, the MD protocol, the classification thresholds, the contig strategy, the filters — each with the options weighed, the choice made, and the check that backs it, mirrored in `docs/DECISIONS.md`.

2 Validating an energetic epitope definition

Before an energetic definition can drive design, its per-residue energies have to be trustworthy, so we test them where the answer is already known. The pilot is 3HFM, the HyHEL-10 Fab bound to hen egg lysozyme, for which SKEMPI records experimental alanine-scanning $\Delta\Delta G$ on thirteen antigen residues. Three dominate — Lys96 (+6.5 kcal/mol), Lys97 (+5.6), Tyr20 (+4.3) — and the rest sit at or below about 1 kcal/mol. We chose 3HFM because it is a well-measured hot-spot system and because lysozyme is already a system with a validated molecular-dynamics setup to build on; none of our own antigen structures appear in SKEMPI.

The method under test is a direct per-residue interaction energy. We run a 20 ns simulation of the solvated complex and, for each antigen residue, average its nonbonded interaction with the antibody (reaction-field Coulomb plus Lennard-Jones) over the trajectory. This is an interaction energy, not a binding free energy: it omits the desolvation a residue pays to make a contact and any entropic contribution. We therefore treat it as a ranking signal and ask one specific question of it.

The question is asymmetric, by design. For design we need to know which residues to fix by identity, so a *false positive* — a residue the energy calls important but that is not — is expensive: we would spend design freedom locking a residue that does not carry the binding. A *false negative* is cheap: a residue that matters through structure, orientation, or entropy, which a per-residue energy cannot see, is exactly the kind we hold by shape rather than identity, a Category-2 residue.

So we do not require the energy to recover every hot spot. We require that everything it flags is real. The criterion is precision, not recall: no residue with high interaction energy may have a low measured $\Delta\Delta G$.

The plan is an empirical fit. We tune the method — the cutoff, which terms enter, whether the solvent channel is included — to empty the false-positive region on 3HFM, then test whether the same method generalizes to other antibody–antigen complexes. If direct interaction energy alone cannot be made free of false positives, that is the evidence that desolvation or water-mediated contributions have to enter, and we move to a fuller treatment.

2.1 What the interaction energy alone recovers

The bare interaction energy recovers the strongest hot spots — Lys97 and Lys96 are its top two antigen calls, and Tyr20 and Asp101 also come out favorable — but it produces a clear false positive at Arg73, which ranks as the second-strongest interaction of all 129 residues yet has a measured $\Delta\Delta G$ of only -0.33 kcal/mol (Figure 1). This is the failure a bare interaction energy is expected to make: Coulomb credits a charged surface residue for its contact without charging it the cost of stripping away the water it was hydrogen-bonded to in solution. Removing the Coulomb term entirely and ranking on Lennard-Jones alone demotes Arg73 and lifts the rank correlation with $\Delta\Delta G$ from 0.46 to 0.73, but it then over-calls well-packed residues that are not hot spots (Leu75) and, at a fixed high-energy cut, multiplies the false-positive count rather than reducing it. Electrostatics over-calls charged residues and packing over-calls buried ones; both point at the same missing quantity, the net free-energy cost of desolvation.

2.2 Adding desolvation: per-residue MM-GBSA

MM-GBSA is the principled fix. It keeps the Coulomb and Lennard-Jones interaction but adds the generalized-Born polar-desolvation term, so a residue's raw contact is netted against the cost of shedding its solvent; run single-trajectory, it decomposes onto the same 20 ns trajectory with no new simulation. On the residue that motivated it, the mechanism works exactly as intended: Arg73's favorable Coulomb contribution (-58.7 kcal/mol) is almost entirely cancelled by the $+57.2$ it pays to desolvate, so its net per-residue contribution falls to -3.6 and it drops from the second-strongest raw call to the flag threshold (Figure 2).

The same trajectory sampled at twenty frames, however, is too sparse for the generalized-Born term to be quantitatively reliable. It over-cancels the largest hot spot: Lys96's Coulomb (-34.0) is overtaken by a $+38.9$ desolvation penalty, leaving it net *unfavorable* ($+2.1$) — a false negative on precisely the residue we most need to recover. The rank correlation collapses to 0.05, and two residues (Arg73, Leu75) still sit marginally inside the false-positive zone. The desolvation term is doing the right physics — it is the only one of the three methods that nets a charged residue's contact against its solvation — but a single 20 ns replica sampled every nanosecond does not give a per-residue generalized-Born estimate stable enough to lock a design against.

2.3 Reading

No method cleanly passes the precision test on 3HFM yet, and the three fail differently (Table 1): the bare interaction energy over-calls charged residues, Lennard-Jones alone over-calls packed ones, and desolvation-aware MM-GBSA corrects the charged over-call in direction but, at this sampling, trades it for noise that sinks the strongest hot spot. The next lever is therefore sampling, not a new estimator: more frames, and likely independent replicas, before the per-residue energy is allowed to classify residues into the three categories. That is a "find-what-works-on-this-system" step, deliberately ahead of the question of what is cheap enough to run at design scale.

Table 1: Per-residue methods on 3HFM against thirteen measured antigen $\Delta\Delta G$. Spearman ρ is over the measured residues; false positives are measured residues above the 90$^{\text{th}}$ interaction-energy percentile with $\Delta\Delta G < 1$ kcal/mol. Reproduce with `energetics/plot_ie_vs_ddg.py` at `--channel` `ab_total`, `ab_lj`, `mmgbsa_dg`.

Method	Spearman ρ	False positives	Characteristic failure
Interaction energy (Coulomb + LJ)	+0.46	4	over-calls charged residues (Arg73)
Lennard-Jones only	+0.73	7	over-calls packed residues (Leu75)
MM-GBSA (20 frames)	+0.05	2	demotes Arg73; over-desolvates Lys96

3 Open questions: entropy and dynamics

(Open thread. No results yet — this records the questions and the formalism to answer them, so the entropy avenue is set down rigorously before we spend the sampling on it.)

Our per-residue interaction energy, and the MM-GBSA that extends it, is an *enthalpic* map of the epitope. The Category-2 residues — those that matter for structure, orientation, or dynamics rather than direct contact energy — are by construction invisible to it. The question is whether the fluctuations in an MD trajectory can supply a per-residue *entropic* signal. This section sets down the relations that would let us, with their assumptions and their failure modes.

3.1 Energy fluctuations give heat capacity, not entropy

In the canonical ensemble ($\beta = 1/k_B T$, $Z = \sum_i e^{-\beta E_i}$), with $\langle E \rangle = -\partial \ln Z / \partial \beta$, the energy variance is

$$\langle \delta E^2 \rangle = \langle E^2 \rangle - \langle E \rangle^2 = \frac{\partial^2 \ln Z}{\partial \beta^2} = k_B T^2 \, C_V. \tag{1}$$

So a single trajectory at one temperature gives the heat capacity $C_V = \langle \delta E^2 \rangle / (k_B T^2)$ exactly — but the entropy needs a temperature integral,

$$S(T) = S(0) + \int_0^T \frac{C_V(T')}{T'} \, dT'. \tag{2}$$

A single-temperature variance is one point on that curve, not S. Raw energy fluctuations therefore do not give entropy without a temperature series. This is the honest baseline against which the next two methods are the exceptions.

3.2 Interaction entropy: entropy from interaction-energy fluctuations

The single-trajectory exception is the interaction-entropy method. Writing the binding free energy from the protein–ligand interaction energy E^{int} and separating its mean (enthalpic) part from the entropic remainder gives

$$-T\Delta S = k_B T \, \ln \left\langle e^{\beta \, \delta E^{\text{int}}} \right\rangle, \qquad \delta E^{\text{int}} = E^{\text{int}} - \langle E^{\text{int}} \rangle, \tag{3}$$

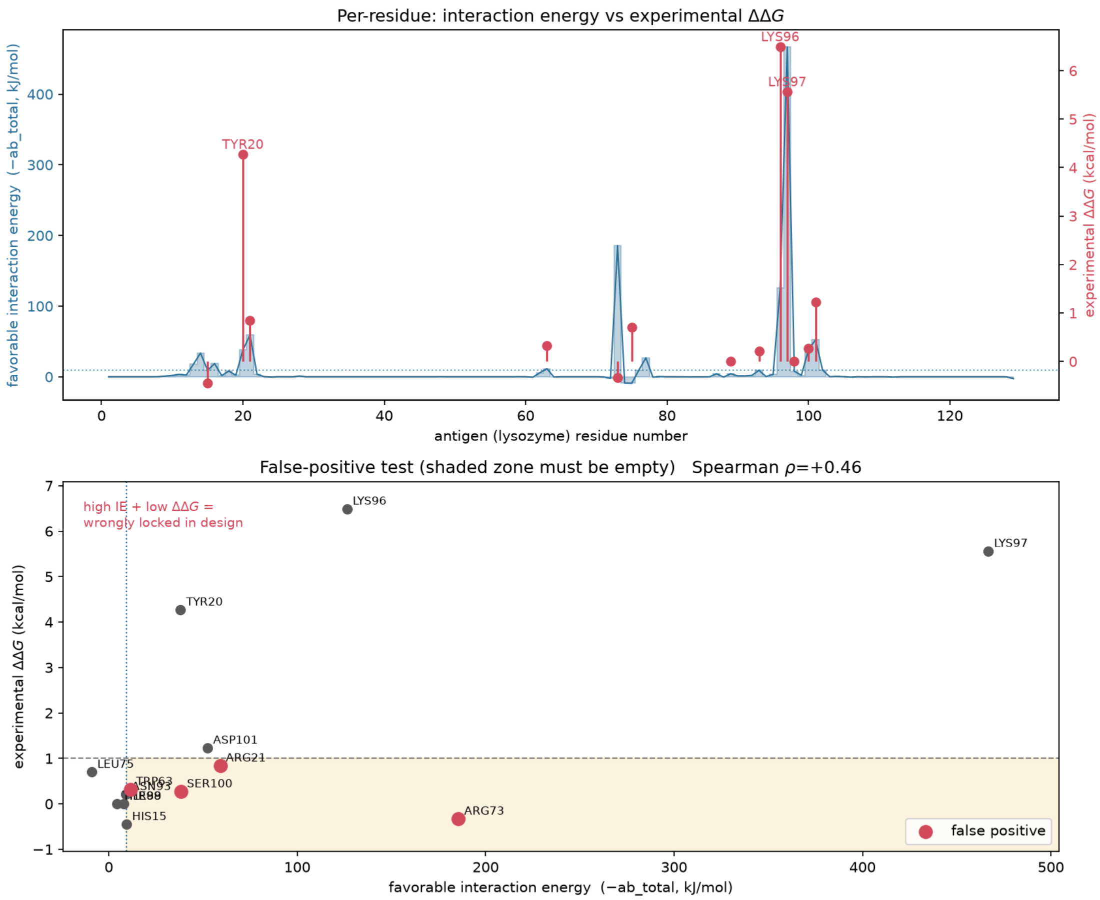

Figure 1: Bare per-residue interaction energy (Coulomb + LJ) against experimental $\Delta\Delta G$ for 3HFM. Lower panel: the false-positive test. The shaded lower-right region — high interaction energy, low measured $\Delta\Delta G$ — must be empty; Arg73 sits inside it, the charged over-call the desolvation step must fix. Generated by `energetics/plot_ie_vs_ddg.py` (`--channel ab_total`) from `holo_ie.py` output.

with the averages over the trajectory. The entropy is encoded in the width and shape of the interaction-energy distribution: a rigid interaction (a delta at $\langle E^{\text{int}}\rangle$) gives $-T\Delta S = 0$; a broad distribution gives a large penalty. Two properties matter here: it is *total* — one number for the whole interface, not per-residue — and the exponential average is dominated by the positive tail of δE^{int} (rare, high-energy configurations), so it converges slowly.

3.3 Configurational entropy: the per-residue route

A per-residue signal must come from *positional* fluctuations. In the quasi-harmonic approximation, form the mass-weighted covariance matrix of the fluctuations, $q = M^{1/2}x$,

$$\sigma_{ij} = \big\langle (q_i - \langle q_i\rangle)(q_j - \langle q_j\rangle)\big\rangle, \tag{4}$$

and diagonalize it. Schlitter's upper bound on the configurational entropy is

$$S_{\text{Schlitter}} = \tfrac{1}{2}k_B \,\ln\det\Big[\,\mathbf{1} + \frac{k_B T\, e^2}{\hbar^2}\,\sigma\,\Big], \tag{5}$$

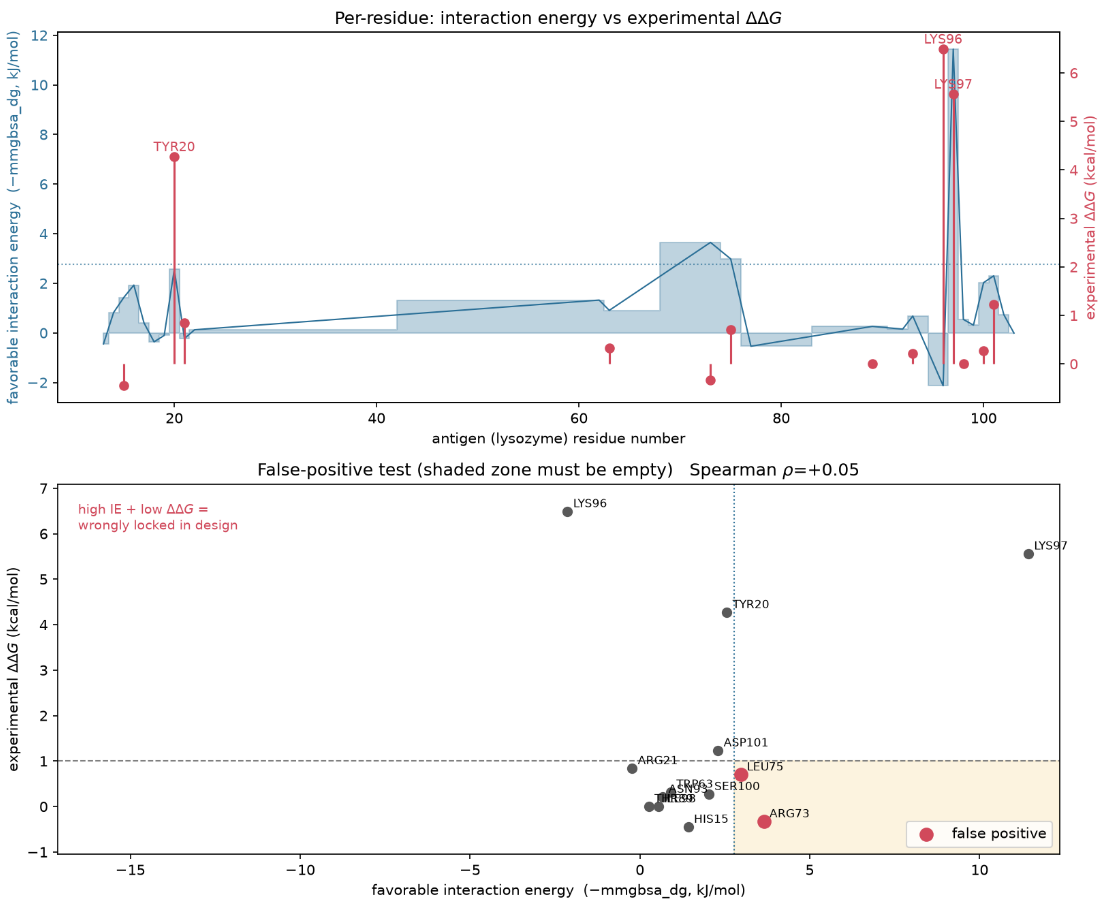

Figure 2: Desolvation-inclusive per-residue MM-GBSA (ΔG) against experimental $\Delta\Delta G$ for 3HFM, same trajectory, 20 frames. Arg73 is pulled back to the threshold (desolvation cancels its Coulomb), but Lys96 — the largest measured hot spot — is over-desolvated to a net-unfavorable ΔG and strands in the upper-left, and the rank correlation collapses ($\rho = 0.05$): the mechanism is right but 20-frame single-replica GB is too noisy. Generated by `energetics/plot_ie_vs_ddg.py` (`--channel mmgbsa_dg`) from `mmgbsa_decomp_to_csv.py` output.

(e Euler's number). The tighter quasi-harmonic form assigns each covariance eigenvalue λ_k an effective frequency $\omega_k = \sqrt{k_B T/\lambda_k}$ and sums harmonic-oscillator entropies,

$$S = k_B \sum_k \left[\frac{\alpha_k}{e^{\alpha_k} - 1} - \ln\!\left(1 - e^{-\alpha_k}\right) \right], \qquad \alpha_k = \frac{\hbar\omega_k}{k_B T}. \tag{6}$$

A per-residue estimate restricts σ to a residue's own atoms, which neglects cross-residue correlations and is therefore approximate and non-additive. The crude scalar proxy is the root-mean-square fluctuation, $\mathrm{RMSF}_i = \sqrt{\langle (x_i - \langle x_i \rangle)^2 \rangle}$; in a harmonic well $S_i \sim k_B \ln \mathrm{RMSF}_i + \text{const}$, so RMSF tracks local entropy monotonically.

3.4 The binding-relevant quantity, and why its sign is ambiguous

What matters for binding is the *change* on binding, which is why the apo simulation is required:

$$\Delta S_i = S_i^{\text{bound}} - S_i^{\text{apo}}. \tag{7}$$

A residue that rigidifies on binding ($\mathrm{RMSF}_i^{\mathrm{bound}} < \mathrm{RMSF}_i^{\mathrm{apo}}$) loses conformational entropy, $-T\Delta S_i > 0$ (unfavorable). But its *contribution to affinity* is not decided by ΔS_i alone:

- **Pre-organization:** a residue already ordered in the apo state pays little entropy on binding — favorable, and the rigid-design intuition (less entropy is better) applies.
- **Functional dynamics / enthalpy–entropy compensation:** retained motion in the bound state can contribute favorable entropy, and rigidification is often repaid by enthalpy. Flexibility is then a feature, not a cost.

The rigorous statement is that the per-residue free energy $\Delta G_i = \Delta H_i - T\Delta S_i$ ranks a residue, and neither term alone fixes the sign. RMSF gives the magnitude of motion, not the sign of its effect on binding.

3.5 Dynamic coupling

Single-residue entropy ignores correlated motion. The coupling between residues i and j can be quantified from the same trajectory by a generalized correlation from mutual information $I(i, j)$,

$$r_{\mathrm{MI}}(i,j) = \left(1 - e^{-2I(i,j)/d}\right)^{1/2}, \tag{8}$$

(d the dimensionality), which captures nonlinear coupling the linear covariance misses. A dynamic-coupling network asks a different, partly sign-free question — is this residue's motion coupled to the interface — rather than whether its entropy is favorable.

3.6 Practical limits, and the design reframe

All of these need long sampling ($\sim 10^2$ ns) and ideally replicates to converge, and per-residue entropy is non-additive because the motions are correlated. So this is a deliberate, single-system investigation, not yet a scalable per-design metric. For the *design* decision it may not even be on the critical path: the Category-2 question — must a residue's shape and orientation be preserved to present the epitope — is closer to a geometric test, which the RFdiffusion backbone-RMSD filter probes directly, than to the full binding thermodynamics. We keep the entropy/dynamics work as mechanistic understanding, separate from the design gate, so the compute is a chosen investment rather than a dependency.

3.7 The concrete open questions

1. Does per-residue ΔRMSF (bound − apo) flag the hot spots the energy map misses (the false negatives, e.g. Tyr20), and does it separate them from the energetic Category-1 residues?
2. Does per-residue quasi-harmonic $-T\Delta S_i$ add anything beyond ΔRMSF, and is it converged at accessible trajectory lengths?
3. Does the total interaction entropy $-T\Delta S$ improve the binding ΔG enough to matter, and does it converge on this system?
4. Is there dynamic coupling (r_{MI}) between epitope residues and the interface that a static picture misses — the specific coupling the rigid paradigm would not predict?

References